Project Sprint #1

The Solitaire game is described in CS449HomeworkOverview. You should read the description very carefully.

In this assignment, you aim to specify the requirements (i.e., user stories and acceptance criteria) of the target software that allows a human player to play a Solitaire game. These requirements will be fully implemented by the end of sprint 3. The minimum features include **choosing the board size and type, starting a new game**, **making a move, determining if a game is over**. The following is a sample GUI layout.

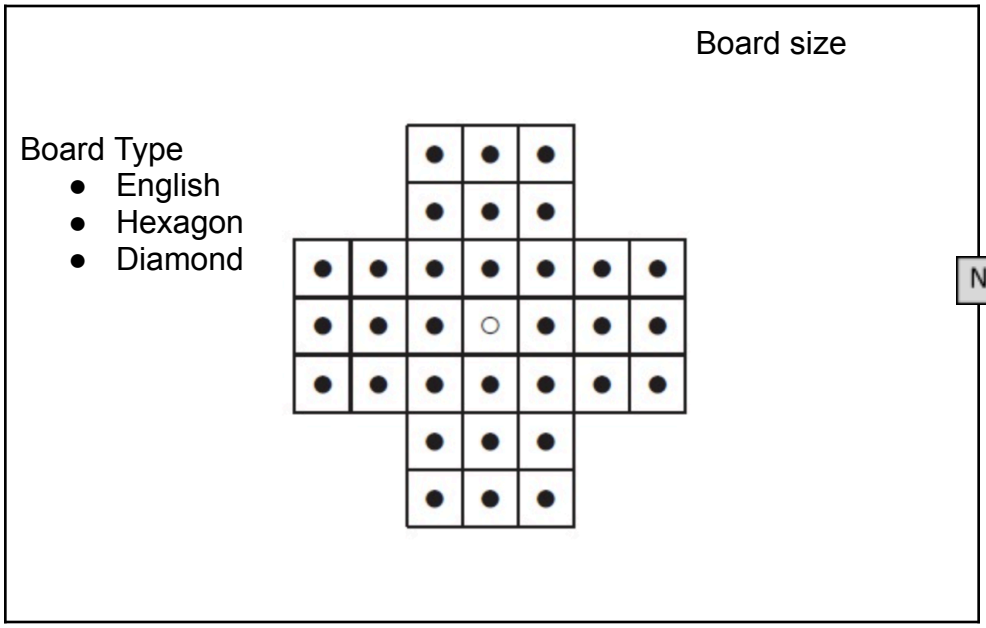


Figure 1. Sample GUI layout of the program by the end of Sprint 3 first working

Use the following tables to ...nt your user stories and acceptance criteria.

You are required to use the free ChatGPT or another LLM to complete two user stories and two of their respective acceptance criteria. You must ensure that the generated user stories are correct and fix them if not. At the end of the submission, provide screenshots of your LLM prompts and answers, along with the description of errors the LLM made that you had to correct. Points will be deducted if no screenshots/description are provided.

For the other user stories and acceptance criteria, you are not required to use LLM.

Note: LLM user stories are highlighted in yellow. Screenshots of prompts and descriptions of errors and corrections will be at the bottom of the main table. I added 3 additional features besides the minimum, which are 6, 7, and 8. I added a section for the autoplay feature (number 9) which is a requirement per the project overview, although it says it is meant for sprint 4 and this is meant to be done by the end of sprint 3, I still made it its own user story as I felt it would be clearer encapsulating into one rather than ingraining it into each user story and for future purposes. Also, I apologize for the formatting at the top (which I hope isn't a big issue), it's that there was a formatting error when I uploaded it into google docs.

I. User Stories (3 points)

- **User Story Template**: As a <role>, I want <goal> [so that <benefit>]

ID	User Story Name	User Story Description	Priority	Estimated effort (hours)
1	Choose a board size	As a peg solitaire player, I want to be able to choose the board size, so that I can play on a board as large or as small as I want.	Medium	3
2	Choose the board type	As a peg solitaire player, I want to choose my board type (English, Hexagon, Diamond) so I can play on my preferred board layout.	Medium	3
3	Start a new game of the chosen board size and type	As a peg solitaire player, I want to start a new game with my desired board size and type. So I can play with what style and size I want.	High	6
4	Make a move in a game	As a peg solitaire player, I want to jump a peg over an adjacent peg into an empty hole, so that the jumped peg is removed and I can work toward clearing the board.	High	8
5	A game is over	As a peg solitaire player, I want the game to tell me when it has ended and how well I did, so that I know when to stop playing and can see my result.	High	5
6	Listening to game music	As a peg solitaire player and music enjoyer, I want to be able to toggle on and off the provided music, so I can play with my preferred music option.	Low	2
7	Choose a theme Note: The theme is just planned to be background, not necessarily a mixture of fonts, music, sound effects, etc.	As a peg solitaire player and design enjoyer, I want to be able to switch between provided themes, so that I can have a different visual experience when I play.	Low	2
8	Show game hints	As a new peg solitaire player, I want the game to be able to show me hints of legal moves, so that I can have assistance for learning to play the game.	Low	4
9	Play in Autoplay	As a peg solitaire player, I want the game to be able to have autoplay, so that I can observe how the game is played.	High	6 (if prior functions work as intended)

Priority Justification:

1. **Medium:** The feature is important to the overall experience (and assignment), but the game can still function if only one size is available
2. **Medium:** As number 1, the feature is important to the experience, but the game can still function with a single board type.
3. **High:** It is imortantt to include the different options and playstyles to have the different experiences, which makes it at least a medium priority, but it also includes that the game must be able to start, which is critical to the function of the program.
4. **High:** Jumping pegs is the core mechanic of peg solitaire. Without it the game cannot be played at all, so it is critical to the function of the program.
5. **High:** Players need to know when no more moves are possible, or they could be stuck on a board with nothing left to do. Detecting the end of a game is part of the core game loop and also connects to the hint behavior in story 8.
6. **Low:** This option is simple for experience and not essential to gameplay or add to the specific mechanics

7. **Low:** Similar to number 6, it adds to the experience but not essential to gameplay
8. **Low:** The game can still be played but it does add a great feature to learning the game and to the experience
9. **High:** Part of the product requirements in the project overview in sprint 4.

Note: I somewhat used peg and marble interchangeable in some instances. They mean the same thing, they mean a piece to be moved. If it an empty space, it is typically referred to as an empty space or empty peg.

II. Acceptance Criteria (AC) (12 points): Add/delete rows as needed.

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inPprogress)
1. Choose a board size	1.1	AC 1.1 Successful pick of a valid board size **Given**: The game board setup screen is displayed with a default board size of 7 **When**: I select an odd whole number between 5 and 9 (5, 7, 9), (e.g: 7) in the board size field **Then**: The board size field shows 7 (or whatever the selected size was) **And** the game will be created with a board size of 7	Completed
	1.2	A.C 1.2 Restriction of Even Board Sizes Given: The game boardsetup screen is displayed When: The userviews the options for the board size Then: The only valid options are 5, 7, and 9 as radio buttons And: Even numbers such as 6 or 8 are excluded from the list	Completed
	1.3	AC 1.3 Rejecting Against Invalid or Out of Range inputs for board size **Given:** The Game board setup screen is displayed **When:** The user interacts with the board size selection **Then:** The UI restricts input strictly through the provided radio buttons, preventing out of range values, decimals, and symbols from being chosen	Completed
2. Choose the board type	2.1	AC 2.1 Default board type and size upon launch of game **Given:** The game has just been launched **When:** The game setup screen is displayed **Then:** The three main board types are offered to pick from (English, Hexagon, Diamond) **And** English is selected by default	Completed
	2.2	AC 2.2 Successful Selection of game board type **Given:** the setup screen is displayed and English is selected (for example) **When:** I select a different board type (example, Hexagon) **Then:** Hexagon becomes the selected board type **And** English is no longer selected **And** The next game I play with be created with the Hexagon board type (applies to other board types, just Hexagon for this scenario)	Completed
	2.3	AC 2.3 Only one board type can be selected at a time **Given:** the setup screen is selected and Diamond is the chosen board type (for example, will apply to other ones) **When:** I select Hexagon (or any other board type) **Then:** only Hexagon is selected (or other chosen board type) **And** the current board size remains unchanged	Completed
3. Start a new game of the chosen board size and type	3.1	AC 3.1 Starting a new English game **Given:** the board size is 7 (default) and the game is set to English **When:** a new game is created by clicking the button **Then:** a cross shaped board with 33 positions are shown And 32 contain a peg And only middle peg space is empty	Completed

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inPprogress)
	3.2	AC 3.2 Creating a game of a different board size and type **Given:** the board type is Diamond and size is 9 (for example) **When:** I click the button to create a new game **Then:** A diamond shaped board with 41 peg positions are shown And 40 contain a peg (for this specific example) And only the center peg space is empty	Completed
	3.3 (No longer applicable, covered under stories 1 and 2)	AC 3.3 Rejecting the ability to start a new game of invalid board size (out of range, even, non numeric character) **Given:** The board field contains an invalid value (as mentioned in header) **When:** I click the button to create a new game **Then:** An error message is displayed (no more error message) And the previous board type remains selected And the board size defaults to 7 Note: This will be removed, but is simply staying here as a reminder.	Complete (no longer applicable as own acceptance criteria as it is too similar to acceptance criteria in stories 1 and 2 to remove redundancy
4. Make a move in a game	4.1	**AC 4.1: Selecting a peg that has a legal move** **Given**: A game is in progress and the board is displayed **When**: I click a peg that has at least one legal jump available **Then**: The peg is shown as selected **And**: Each empty hole it can legally jump into is highlighted	Completed
	4.2	**AC 4.2: Making a valid jump** **Given**: A peg is selected and a highlighted empty hole is two positions away in a straight line (either diagonal or orthogonal), with a peg between them **When**: I click that empty hole **Then**: The selected peg moves into the empty hole **And**: The peg that was jumped over is removed from the board **And**: The hole the peg started in becomes empty **And**: The total number of pegs on the board decreases by one	Completed
	4.3	**AC 4.3: Rejecting an illegal destination** **Given**: A peg is selected **When**: I click a hole or position that is not a legal jump **Then**: No move is made and the board stays unchanged **And**: A message is displayed indicating that the move is not allowed **And**: The peg becomes unselected	Completed
	4.4	**AC 4.4: Rejecting selection of an empty hole or a peg with no legal moves** **Given**: A game is in progress **When**: I click an empty hole as the starting point, or a peg that has no legal jumps **Then**: Nothing is selected and the board stays unchanged **And**: A message is displayed indicating that there is no move available from that position	Completed
	4.5	**AC 4.5: Deselecting a peg** **Given**: A peg is selected **When**: I click the selected peg again **Then**: The peg is no longer selected **And**: The destination highlights disappear	Completed

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inPprogress)
		And: The board stays unchanged	
	4.6	**AC 4.6: Selecting another legal peg when one is currently selected** **Given:** A peg is selected **When:** I click another peg with a legal move **Then:** The new peg is highlighted along with its legal moves And: The previously peg that was selected no longer shows it as selected And: The previous peg's legal moves are no longer shown	Completed
5. A game is over	5.1	**AC 5.1: Game ends when no legal moves remain** Given: A game is in progress and more than one peg remains on the board When: I make a move that leaves no peg with a legal jump Then: A message is displayed saying the game is over because no legal moves remain And: The number of pegs remaining is displayed And: I can select an option to create a new game (either restart current one to play the same mode, or press back button to select a different one) And: I can see my game performance, Very Good (2 marbles left), Good (3 marbles left), Average (4 or more left).)	Completed
	5.2	**AC 5.2: Winning the game** Given: A game is in progress When: I make a move that leaves exactly one peg on the board Then: A message is displayed saying that I won And: The final peg count of 1 is shown And: I can select an option to create a new game And: I can see my rating of Outstanding (1 marble left)	Completed
	5.3	**AC 5.3: Game continues while legal moves exist** Given: A game is in progress When: I make a move and at least one peg still has a legal jump Then: The game does not end And: No game over message is displayed	Completed
	5.4	**AC 5.4: Moves are disabled after the game ends** Given: The game over message is displayed When: I click a peg or an empty hole on the board Then: No peg is selected and no move is made And: The board stays as it was when the game ended And I can see my game rating (Outstanding (1 marbleleft), Very Good (2 marbles left), Good (3 marbles left), Average (4 or more left).	Completed
	5.5	**AC 5.5: Starting a new game after game over** **Given: The game over message is displayed** **When: I click the button to start a new game (via restart or going back)** **Then: A new board is created using the selected board type and size** **And: The game over message is cleared** **And: I can play the new game that was created (via restart or back)**	Completed
6. Enabling/Disabling Music	6.1	AC 6.1 Default playing of game music **Given:** The game has been launched **When:** I see the opening screen **Then:** I should be able to hear the provided game music And a button with a speaker icon indicating that music is enabled appears	toDo

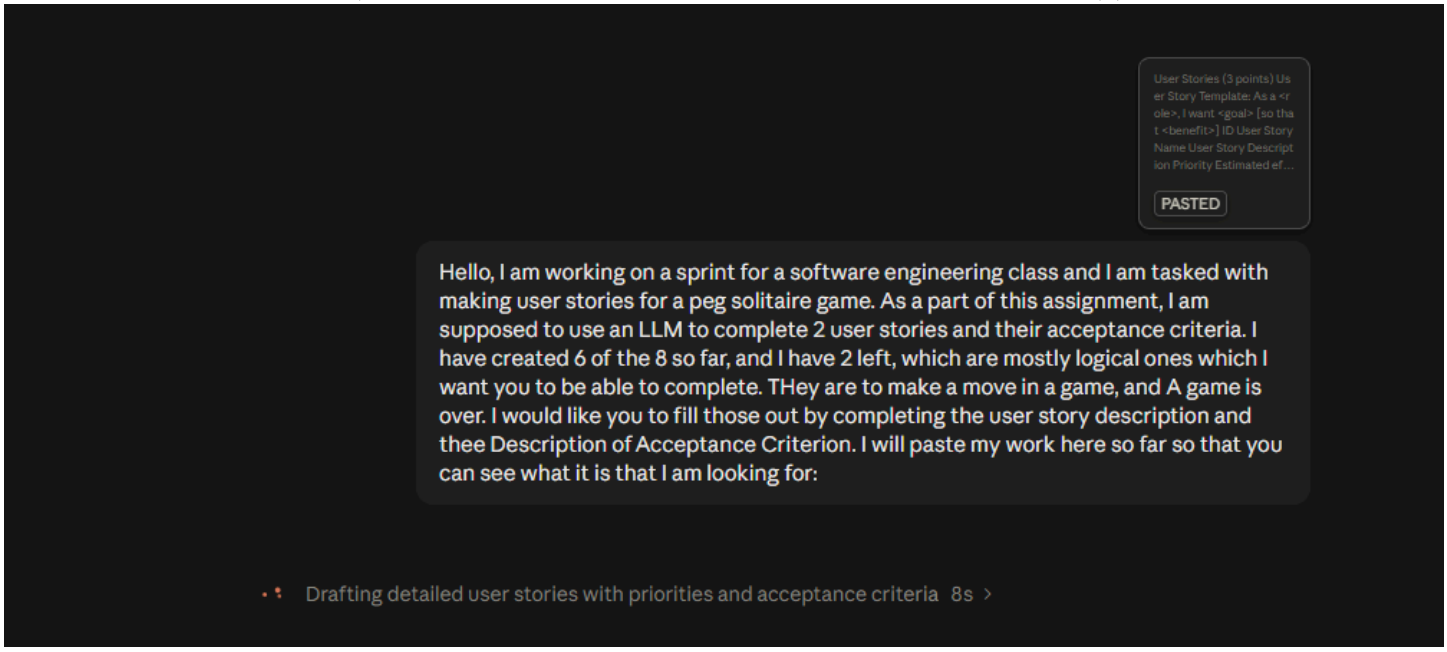
User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inPprogress)
	6.2	AC 6.2: Turning off the music **Given:** The game music is playing **When:** The toggle button has the icon to show that music is playing **Then:** The music should turn off when the button is clicked And: The toggle button icon should display an X on it, indicating it is off	toDo
	6.3	AC 6.3: Turning on the music **Given**: The game music is off **When**: the toggle button has the icon to show that music is not playing **Then:** The music plays And it starts at the beginning of the soundtrack	toDo
	6.4	AC 6.4: The soundtrack ends **Given**: The game music is playing **When:** The length of the soundtrack finishes **Then**: The music soundtrack should restart from the beginning (not be silent)	toDo
7. Choose a theme	7.1	AC 7.1: Default theme is shown upon launch **Given:** The game is launched **When:** The game opening screen is displayed **Then:** The default theme should be displayed The default theme button should be selected And the other themes should not be selected	toDo
	7.2	AC 7.2: Changing theme **Given:** The game is open in any screen **When:** The button for a different theme than the current one is clicked **Then:** The background theme should change to the corresponding selected them And the button should be shown as selected And the previous theme button should not be selected And other theme buttons should not be selected	toDo
8. Give move hints	8.1	AC 8.1: Clicking hints button to show hints during gameplay **Given:** A game has started and the board is displayed **When:** I click the button to show hints **Then:** A colored ring will appear around each and every peg that has a legal move And: There will be a message displaying and number of how many pegs can be moved	toDo
	8.2	AC 8.2: Clicking hints when there are no more legal moves **Given:** A game has started and the board is displayed **When:** I click the hint button and there are no more legal moves **Then:** A message is displayed saying that there are no more legal moves and that this game is over	toDo
9. Play the game in Autoplay	9.1	AC 9.1: Starting autoplay **Given:** a game is in progress and there is at least one legal move **When:** I click the autoplay button **Then:** The computer makes moves on the board without user input **And** each move is shown on the board as it is made (pegs jump and disappears and moving peg lands in empty hole as appropriate)	toDo
	9.2	AC 9.2: Autoplay moves are randomized and legal **Given:** Autoplay is running **When:** the computer chooses its next move **Then:** The move is a legal move (orthogonal or diagonal over adjacent peg into empty hole two positions away - description in project overview)	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inPprogress)
		And: The move is picked at random from all available legal moves (not hardcoded)	
	9.3	AC 9.3: Autoplay ends with no legal moves remaining **Given:** Autoplay is running **When:** a move leaves with no more legal moves available **Then:** Autoplay stops with a game over message displayed And the number of pegs remaining is shown And the performance rating is displayed (Good, Avg. etc.) And I can select an option to create a new game	toDo
	9.4	AC 9.4: Manual input is blocked/not allowed while autoplay is on **Given:** Autoplay is running **When:** I click a peg or empty hole on the board **Then:** No peg is selected and no move is played And autoplay continues as normal (unaffected)	toDo
	9.5	AC 9.5: Autoplay works on every board type and size **Given:** A new game is created with the supported/available board types (English, hexagon, Diamond) and valid board size (5,7,9) **When:** I click to enable autoplay **Then:** Autoplay plays the game to completion following the same rules as manual play would on the board	toDo
	9.6	AC 9.6: Starting a new game after autoplay **Given:** An autoplay game has ended with game over message **When:** I click o start a new game **Then:** A new board is created with designated type and size And The game over message is cleared And I can play manually or in autoplay	toDo

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LLM: Claude

Prompts for User Story 4 (pasted input is my work for stories 1,2,3,6,7, and 8 (I added 9 after I did the LLM ones)):



Output:

User Story 4: Make a move in a game

Field	Content
ID	4
Name	Make a move in a game
Description	As a peg solitaire player, I want to jump a peg over an adjacent peg into an empty hole, so that the jumped peg is removed and I can work toward clearing the board.
Priority	High
Estimated effort	8 hours

Priority justification: High: Jumping pegs is the core mechanic of peg solitaire. Without it the game cannot be played at all, so it is critical to the function of the program.

Acceptance Criteria

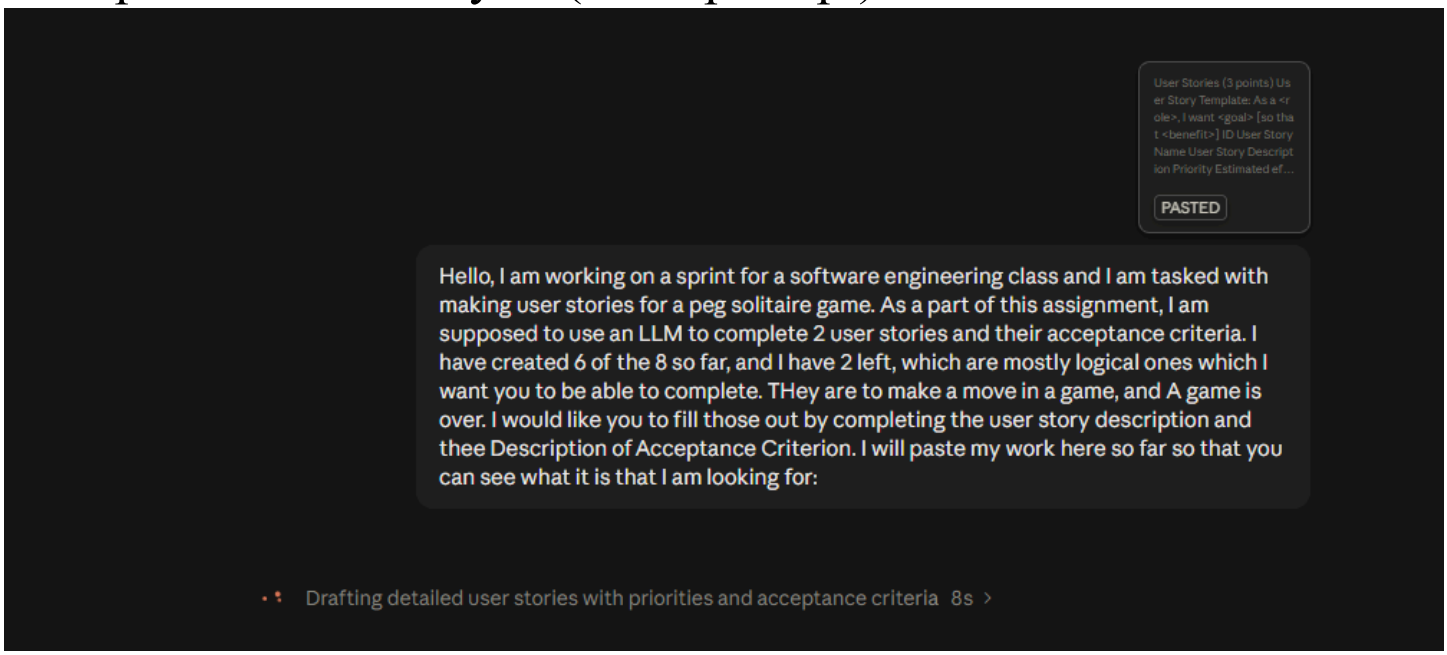
AC ID	Description	Status
4.1	**AC 4.1: Selecting a peg that has a legal move** Given: A game is in progress and the board is displayed When: I click a peg that has at least one legal jump available Then: The peg is shown as selected And: Each empty hole it can legally jump into is highlighted	toDo
4.2	**AC 4.2: Making a valid jump** Given: A peg is selected and a highlighted empty hole is two positions away in a straight line, with a peg between them When: I click that empty hole Then: The selected peg moves into the empty hole And: The peg that was jumped over is removed from the board And: The hole the peg started in becomes empty And: The total number of pegs on the board decreases by one	toDo
4.3	**AC 4.3: Rejecting an illegal destination** Given: A peg is selected When: I click a hole or position that is not a legal jump (for example a diagonal position, a hole only one space away, a hole with no peg between, or a position occupied by another peg) Then: No move is made and the board stays unchanged And: A message is displayed indicating that the move is not allowed And: The peg remains selected	toDo
4.4	**AC 4.4: Rejecting selection of an empty hole or a peg with no legal moves** Given: A game is in progress When: I click an empty hole as the starting point, or a peg that has no legal jumps Then: Nothing is selected and the board stays unchanged And: A message is displayed indicating that there is no move available from that position	toDo
4.5	**AC 4.5: Deselecting a peg** Given: A peg is selected When: I click the selected peg again Then: The peg is no longer selected And: The destination highlights disappear And: The board stays unchanged	toDo

Descriptions of Errors Made and Corrections for User Story 4:

Overall, I mostly agree with Claude with the user story it made and its criteria. I believe in its estimated work time, priority, and overall description of each point. The language it used is clear and gives examples when necessary. I do however believe it did miss an important criteria, which I highlighted in green at the top as 4.6. It is a combination of 4.5 and 4.2 (slight overlap), but I believe it is important enough to distinguish this possibility

as it would be common and its possible errors could occur. What I did was add a case where a user selects one peg with a legal move, but then deselects it by clicking another legal peg, and not deselecting the current one by clicking it again. Additionally, I specified this one as only another legal move as I believe 4.4 captures the behavior of selecting a non-legal move peg quite well (to avoid overlap). It also mentioned diagonal moves being illegal overall, but the project overview doc mentions diagonal moves so it was removed.

Prompts for User Story 5: (same prompt)



Output:

User Story 5: A game is over

Field	Content
ID	5
Name	A game is over
Description	As a peg solitaire player, I want the game to tell me when it has ended and how well I did, so that I know when to stop playing and can see my result.
Priority	High
Estimated effort	5 hours

Priority justification: High: Players need to know when no more moves are possible, or they could be stuck on a board with nothing left to do. Detecting the end of a game is part of the core game loop and also connects to the hint behavior in story 8.

Acceptance Criteria

AC ID	Description	Status
5.1	**AC 5.1: Game ends when no legal moves remain** Given: A game is in progress and more than one peg remains on the board When: I make a move that leaves no peg with a legal jump Then: A message is displayed saying the game is over because no legal moves remain And: The number of pegs remaining is displayed	toDo
5.2	**AC 5.2: Winning the game** Given: A game is in progress When: I make a move that leaves exactly one peg on the board Then: A message is displayed saying that I won And: The final peg count of 1 is shown	toDo
5.3	**AC 5.3: Perfect win in the center** Given: A game is in progress and the game ends with exactly one peg left When: That last peg is in the center position of the board Then: A message is displayed indicating a perfect win	toDo
5.4	**AC 5.4: Game continues while legal moves exist** Given: A game is in progress When: I make a move and at least one peg still has a legal jump Then: The game does not end And: No game over message is displayed	toDo
5.5	**AC 5.5: Moves are disabled after the game ends** Given: The game over message is displayed When: I click a peg or an empty hole on the board Then: No peg is selected and no move is made And: The board stays as it was when the game ended	toDo
5.6	**AC 5.6: Starting a new game after game over** Given: The game over message is displayed When: I click the button to start a new game Then: A new board is created using the selected board type and size And: The game over message is cleared	toDo

Descriptions of Errors Made and Corrections for User Story 5:

The created user story and criteria created and given to me by Claude do a good job at encapsulating the overall criteria for when a game is won or it ends. An addition that I

made to essentially all the entries was to allow a new game to be created at the end. I felt it was important to specify as it is important to allow the user to continue to use the game and play more. The specifications provided by Claude weren't incorrect, but I felt it was important to specify as it is possible the requirements are all met but one can't play a new game, which is important to have. I also removed the perfect win in the center criteria. Claude provided a criteria where it says if I get a perfect win by having it located in the center then it displays a special message. I removed it, not necessarily because it was wrong but because I simply don't want to include it in my game. Lastly, the next fix I added was to show the more specific performance as provided in the project overview sheet. Claude was not provided this sheet in the prompt so it had no way of knowing, so I added it to align more closely with the requirements. My additions are highlighted in green and the original 5.3 provided by Claude was removed, and following requirements were moved back down by .1.